MILITARY SPECIFICATION

PRIMER, PERCUSSION, M100
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by U.S. Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of percussion primer designated as Primer, Percussion, M100.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-P-20449	- Primer, Percussion, M61, Loading, Assembling and Packing
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

DRAWINGS

ARMY

9277103	- Primer, Percussion M100, Loading Assembly
9328574	- Box, Fiberboard w/Partition Loose
9328575	- Packing and Marking for Box, Wirebound, Packing, for Primer, Percussion, M100 Assembly

FSC: 1390

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement and functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Percussion primer.-The percussion primer shall comply with all requirements specified on Drawing (dwg.) 9277103, and with all requirements specified in applicable specifications.

3.3 Primer assembly

3.3.1 Primer assembly without black powder charge

3.3.1.1 Non-functioning.-The primer assembly (without black powder charge) shall not function under an applied energy value of 47.52 inch-ounces, minimum.

3.3.1.2 Functioning.-The primer assembly (without black powder charge) shall function under an applied energy value of 159.20 inch-ounces, maximum.

3.3.2 Primer assembly with black powder charge

3.3.2.1 Functioning.-If the percussion element fires, the primer assemblies shall function without audible hangfire.

3.4 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time of loading shall not exceed 0.5 percent, when tested as specified in 4.3.2.

3.5 Workmanship.-All parts shall be free of burrs, chips, dirt, grease, cracks, rust corrosion and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

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4.2 Quality Conformance Inspection

4.2.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of primers shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. Black powder of one lot number from one manufacturer.

c. Primers, Percussion, M61 of one lot interfix number from one manufacturer.

4.2.2 Examination.-See MIL-A-48078

a. Sampling plans-Unless otherwise specified in the classification of Defects and Test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II, except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.1	Primer, prior to assembling of battery cup and firing plug		9277103
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Percussion element missing, cocked or inverted		100%	3.2	Visual
Minor					
201	Seal between percussion element assembly and head assembly missing or incomplete		.65%	3.2	Visual
202	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.2	Primer, prior to assembly of liner		9277103
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Firing plug or battery cup above flush with surface of head		100%	3.2	Straight-edge
2	Any flash hole missing from body		100%	3.2	Gage
3	Primer assembly non-functioning		4.2.3.1.1	3.3.1.1	Test/4.3.1.1.1
Major					
101	Firing plug missing, cocked or inverted		.40%	3.2	Visual
102	Flash hole at base of charge cavity obstructed		.40%	3.2	Visual
103	Battery cup loose or missing		.40%	3.2	Visual/Manual
104	Primer assembly functioning		4.2.3.1.2	3.3.1.2	Test/4.3.1.1.2
Minor					
201	Evidence of poor workmanship		.65%	3.5	Visual
202	Roll length of liner, min. (prior to assembly)		.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63210 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9277103
4.2.2.3	Primer, prior to assembly of plug				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of charge		.40%	3.2	Balance
102	Liner fails to cover all holes		.40%	3.2	Visual
103	Moisture content of black powder		See 4.2.3.4	3.4	4.3.2
Minor					
201	Foreign matter in charge		.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.4	Primer percussion assembly				9277103
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any flash hole not completely sealed (Fig. 1)		100%	3.2	Visual
102	Threads of closing plug not coated with sealant		.40%	3.2	Visual
103	Closing plug above flush		.40%	3.2	Straight-edge
104	Closing plug fails to withstand min. disassembly torque		.40%	3.2	Test
105	Primer assembly functioning	25		3.3.2.1	Test/4.3.1.2.1
Minor					
201	Marking incorrect or unidentifiable		.65%	3.2	Visual
202	Evidence of poor workmanship		.65%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.5	Fiberboard box, prior to sealing				9328574
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any packing material missing		.65%	3.2	Visual
202	Separator damaged to extent that primers touch		.65%	3.2	Visual
203	Primer missing		.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9328574
4.2.2.6	Fiberboard box, sealed				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton		.65%	3.2	Visual
202	Carton torn, cut or punctured		.65%	3.2	Visual
203	Marking misleading or unidentifiable		.65%	3.2	Visual
204	Head end of carton not identifiable		.65%	3.2	Visual
205	Contents loose		.65%	3.2	Manual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9328573
4.2.2.7	Sealed, barrier bag				NEXT HIGHER ASSEMBLY 9328575
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Seal improper or incomplete		.40%	3.2	Visual
102	Bag punctured, torn or cut		.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.8	Packing box, prior to sealing				9328575
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Carton missing		40%	3.2	Visual
Minor	None defined				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63210 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.9	Packing box sealed		9328575
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken, board loose or protruding nail		.40%	3.2	Visual
102	Hardware or strapping missing, broken or loose		.40%	3.2	Visual
Minor					
201	Cap seal missing, unsealed or improperly positioned		.65%	3.2	Visual
202	Hardware or strapping improperly engaged or mislocated		.65%	3.2	Visual
203	Marking misleading or unidentifiable		.65%	3.2	Visual
204	Contents loose		.65%	3.2	Manual

NOTES:

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4.2.3 Testing

4.2.3.1 Primer assembly without black powder charge

4.2.3.1.1 Non-functioning.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 150 assemblies shall be randomly selected from each lot for the test. If any sample fails to comply with requirement the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.1. Primers subjected to this test shall be destroyed.

4.2.3.1.1.1 After three consecutive lots have met the criteria of 4.2.3.1.1, sample of 50 assemblies shall be randomly selected from each lot for this test. If one or more primers fail to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.1 using equipment in accordance with 4.2.4. Primers subjected to this test shall be destroyed.

4.2.3.1.2 Functioning.-Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more primers fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found, a second sample of five hundred primers shall be tested. If the combined number of defectives in both the first and second sample is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.2 using equipment in accordance with 4.2.4.

4.2.3.1.2.1 After three consecutive lots have met the criteria of 4.2.3.1.2, a sample of 125 primers shall be randomly selected from each lot for this test. If two or more primers fail to comply with the applicable requirement, the lot shall be rejected. If only one defect is found, a second sample of 250 primers shall be tested. If the combined number of defectives in both the first and second sample is two or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.2 using equipment in accordance with 4.2.4.

4.2.3.2 Primer assembly with black powder charge.-A sample of 25 primers shall be selected from each lot for this test. If any primer fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2.1.

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4.2.3.3 Check test for deterioration of primer, percussion (M61 assembly).-If the total elapsed time between original acceptance of the M61 primer lot and the assembly of that lot into the M100 primer exceeds two years, or if the M61 primers have been subjected to adverse conditions for any period of time since previous test, the M61 primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-20449 immediately before the M61 primer lot is assembled into the M100 primers (see 6.3).

4.2.3.4 Moisture content of black powder.-A sample of one primer shall be randomly selected from each eight hours production. The black powder shall be tested as specified in 4.3.2 and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the paragraph Reference/Inspection in Method column in tables starting with paragraph 4.2.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078.

4.3 Test methods and procedures

4.3.1 Primer assembly

4.3.1.1 Primer assembly without black powder charge

4.3.1.1.1 Non-functioning.-The primer assembly (without black powder charge) shall be assembled into a test fixture (see 4.2.4). A steel ball weighing a minimum of 7.92 ounces shall be dropped from a height to give the required impact force (approximately 6 inches).

4.3.1.1.2 Functioning.-The test shall be performed as specified in 4.3.1.1.1 except that the weight of the ball shall be 7.96 ounces maximum, and the height shall be adjusted to give the required impact force (approximately 20 inches).

4.3.1.2 Primer assembly with black powder charge

4.3.1.2.1 Functioning.-The primers shall be assembled to the complete round for which the primer was designed and fired from the gun for which the round is standard. The primer shall be observed to determine compliance with the requirement.

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4.3.2 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The primers shall be preserved and packaged in accordance with dwg. 9328574.

5.2 Packing

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 9328575.

5.3 Marking.-Marking shall be in accordance with dwgs. 9328574 and 9328575.

6. NOTES

6.1 Ordering data.-See MIL-A-48078.

6.2 Submission of design for approval.-See 6.2.3 of MIL-A-48078. Submit equipment designs as required, to Commander, ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.3 Cost of check tests.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost of the contractor who loaded the M61 primer or the contractor loading and assembling the M100 primer, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., while the primers were under the jurisdiction of either contractor.

6.4 Check test for possible deterioration of percussion artillery primers.-The following tests are designed to prevent the loading of artillery primers which have deteriorated into a major ammunition item. The test shall be performed on artillery primers from each lot which has been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check test in the contract.

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6.4.1 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the body by any suitable means. **This moisture content of the powder shall be determined in accordance with the methods specified in Specification MIL-P-223. If the moisture content of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

6.4.2 Primer head assembly function.-The sampling plan for this test shall be in accordance with Table I. Any primer head assembly which fails to comply with the applicable requirement shall be rejected. The black powder shall be removed from the charge cavity prior to performance of this test.

TABLE I

Sample	Sample Size	Size	Combined Samples Acceptance Number	Combined Samples Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	4	5

* No acceptance on first sample.
** Since black powder is hygroscopic, it is advisable to place the powder in a dry, air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1390-A168

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ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

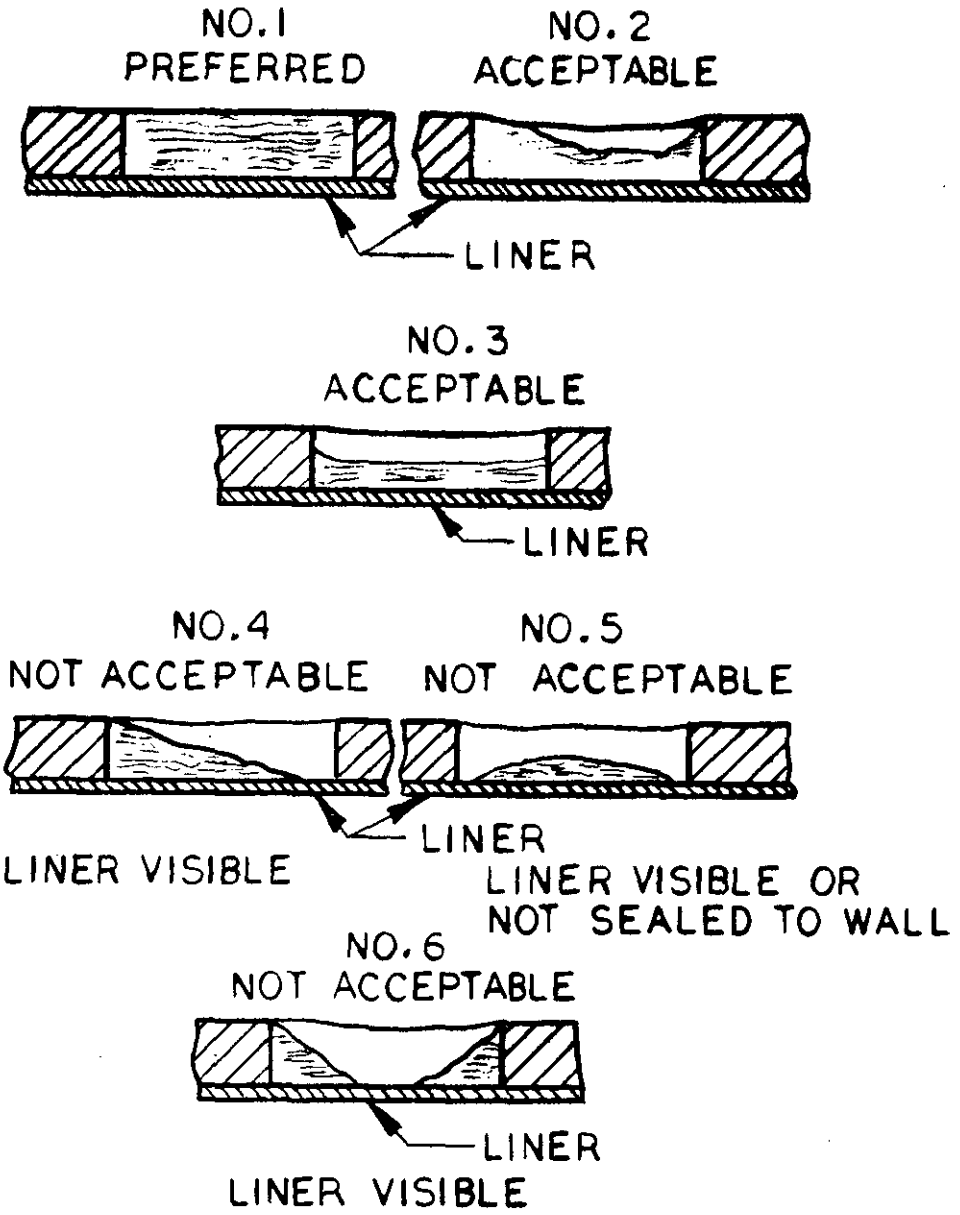


FIGURE 1

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63210 (AR) Primer, Percussion, M100 Loading, Assembling And Packing

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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